Project Dashboard -Altimeter Board

M. Zaeem | Electronics & Embedded Systems Engineer | BSc Electrical Engineering | MSc Autonomous Systems

1. Project Basic Information

Project Name:	Altimeter Board (To record altitude and acceleration)		
Client Name:	Chris Flanigan		
Start Date:	29-Sep-25	**Expected End Date:**	31-Oct-25
Pricing Model:	**Fixed Price** ☐ **Hourly Rate** ☑		
Hourly Rate Per Hour ($):	€ 15.00	**Total Project Budget:**	€ 1,500.00
Payment by Milestone:	☐		

2. Project Requirements

Project Requirements

The project involves designing and building a miniature, altimeter for model rockets. The device must record altitude and optional acceleration in non-volatile memory and allow post-flight data download to a laptop. The system should use Arduino-compatible components and software, with the option for wireless data transfer. The final design will include a lightweight custom PCB fitting within a 17.5 mm × 30 mm cylinder.

Core Requirements:

Microcontroller: nRF52840, Arduino-compatible.
Sensors: Pressure sensor: BMP390, Accelerometer: BMA253 or BMI270 (optional 3- or 6-axis)
Memory: EEPROM for non-volatile storage.
Connectivity: USB-C; optional Bluetooth for data transfer.
Power: Connector for single-cell LiPo battery.
Data Handling: Store flight data in EEPROM and provide access for download to a laptop. Optionally send data via Bluetooth or Arduino Cloud.
PCB Design: Schematic and layout in KiCAD or similar; target mass ≤ 2.5 grams.
Software Platform: Arduino IDE; optional Python/Processing for data analysis.
Hardware Sourcing: Use readily available, reliable components.
Developer Expertise Required: Firmware development, microcontrollers, programming.

3. Project Milestones

	Task Requirements	Hours Spent	Description	Status
Milestone 1- (Hardware selection, component validation, preliminary system design)	Define system architecture and finalize MCU selection	6	System Architecture & Component Selection	Completed
	Identify and validate sensors	2		Completed
	Feasibility analysis of components and connectivity	2		Completed
	Order pre-assembled modules	1		Completed
	Develop basic Arduino firmware for sensor interfacing.	5		Completed
	Milestone 1 Hours :	16		
Milestone 2- (Sensor integration, firmware development, ground station interface)	**Task Requirements**	**Hours Spent**	**Description**	**Status**
	Integrate sensors with MCU and implement non-volatile memory logging	26	Data Acquisition & Storage Implementation	Completed
	Develop Python-based ground station interface for data retrieval via BLE	16		Completed
	Perform functional verification of sensor readout and data logging	4		In Progress
	Benchmark and validate data integrity and transfer reliability	4		Completed
	Implement X-Y plotting of data using Graph.	4		Completed
	Milestone 2 Hours :	54		
Milestone 3- (PCB design, fabrication prep, final delivery)	**Task Requirements**	**Hours Spent**	**Description**	**Status**
	Making detailed schematic diagram (Altium)	16	PCB Design & BOM	Completed
	PCB layout (Altium)			Pending
	Generate Gerber files and assembly documentation			Pending
	Ordering PCBS			Pending
	Uploading Firmware and Bootloader			Pending
	Testing and Pakaging			Pending
	Shipping to Client			Pending
	Milestone 3 Hours :	16		

4. Key Deliverables

		Additional Requirements	Project End Goal and Approval Criteria
Sensor Selection	☑	**Hardware**	All hardware components functioning to specification
Firmware Development	☑	Target Weight ≤ 2.5 grams	Software meets All requirements with no critical bugs
Schematic Drawing	☑	Max Board Size (17.5 mm × 30 mm)	Client signs off on final deliverables
PCB Layout	☑	Powered by LiPo battery, 3.7 V	
Gerber Files	☑	USB-C connector	
Bill of Materials (BOM)	☑	**Software**	**Design**
PCB Fabrication Ordering	☐	Arduino IDE Compatable	User interface
PCB Assembly (soldering components)	☐	EEPROM Data Storage	Data processing
Board Testing	☐	Bluetooth For Data Transfer.	Communication protocols
3D Enclosure Design	☑		Error handling

5. Unallocated Task Time Tracking

Task Description	Hours Spent
Hours:	0

6. Additional Expenses :

Item / Description	Category	Notes	Cost per Unit (€)	Units	Total (€)
PCB Fabrication (JLCPCB)	Fabrication				€ 0.00
Customs VAT	Customs/Tax				€ 0.00

Enclosure Printing	Outsourcing				€ 0.00
PCB Assembly	Assembly Service				€ 0.00
Components	Components LCSC				€ 0.00
Unit Testing Labor	Testing				€ 0.00
Delivery to Client	Logistics				€ 0.00
Others					€ 0.00
				Total Cost:	**€ 0.00**

7. Payment Section

Hourly Payments

Task / Milestone	**Rate (€/hr)**	**Hours Worked**	**Amount (€)**	**Paid (€)**	**Remaining (€)**
Milestone 1	€ 15.00	16	€ 240.00	€ 171.00	€ 69.00
Milestone 2	€ 15.00	54	€ 810.00	€ 429.00	€ 381.00
Milestone 3	€ 15.00	16	€ 240.00	€ 429.00	-€ 189.00
Extra Tasks (Unallocated)	€ 15.00	0	€ 0.00		€ 0.00
Additional Expenses	Nil	Nil	€ 0.00		€ 0.00
Total Amount:		**86**	**€ 1,290.00**	**€ 1,029.00**	**€ 261.00**

Thank you for choosing to work with me. I truly appreciate your trust, and I'll do my best to bring your idea to life with precision, dedication, and transparency. This spreadsheet has been created to provide a clear overview of project scope, progress, and costs, ensuring full visibility at every step.

Please Note:

The quotation provided does not include external expenses such as:
- Component sourcing (AliExpress, Digi-Key, etc.)
- PCB fabrication or assembly services (e.g., JLCPCB, PCBWay)
- Import duties, VAT, or customs clearance fees
- Packaging or shipping of prototypes
- Specialized tooling or any outsourced service costs
- These costs will be discussed and billed separately if required during the project.

Looking forward to building something great together from concept to completion, I'm here as your technical partner.

Warm regards,
M. Zaeem
Electronics & Embedded Systems Engineer